

DeepSequence: Hierarchical Attention Forecaster

DS + Softsign + Monotone Experts + Context-Aware Mixer

1. INPUTS

Heterogeneous feature groups



2. EXPERTS

Specialized monotone expert networks



3. INTRA-EXPERT ATTENTION

Capture structure within each expert



4. HIERARCHICAL MIXER

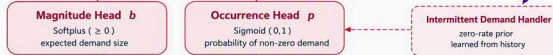
Context-aware mixing across experts



5. FORECAST HEADS

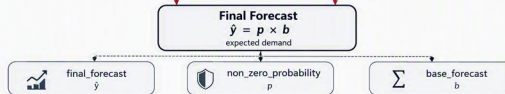
(Gated Heads)

Decompose forecast into magnitude and occurrence



6. OUTPUTS

Final probabilistic forecast



Legend

Inputs

Experts & Intra-Expert

Mixer & Aggregation

Forecast Heads

Outputs

SKU Conditioning (conditioning path)

Optional / Ablation Path

Notation

E_i output of expert i

w_i attention weight for expert i

b magnitude (base forecast)

p non-zero probability

$\hat{y}$ final forecast

Notes

$\dagger$ Daily: lags 1,2,7

Monthly: lags 1,2,12

Quarterly: lags 1,4

Softsign: bounded & stable.

Monotone experts: interpretable effects.

DeepSequence decomposes demand into occurrence (p) and magnitude (b), combines specialized monotone experts with hierarchical attention mixing, and produces robust forecasts for intermittent and non-intermittent SKUs.